

The Atlas Issue — Artometrics

The Atlas Issue

Soft power, rival cities, export identity — culture on a map.

When Films, Music, and Games Become National Exports

Culture is an export category even when it never passes through a port. A song, show, game, cuisine, fashion house, or sports star can change what a country means in the global imagination. This framing highlights **8** cultural-export systems, scores **8** mediums for exportability, and tracks **3** Asian cultural-wave paths — with World Cities Culture Forum summaries covering **45** global cities as context.

The charts use editorial indices to define the cultural-trade question before deeper joins to UNESCO, WIPO, and platform data: what travels, where it is produced, who monetizes it, and how it becomes identity.

Fast facts

The numbers that set the scale for this report:

8Countries and cultural-export systems highlighted

8Mediums scored for exportability

6Diagnostic cultural trade questions

45Global cities in World Cities Culture Forum 5th edition summaries

3Asian cultural wave paths compared

Data and method

The source stack includes UNESCO cultural statistics, WIPO creative-industry indicators, World Bank and OECD services data, MusicBrainz, IMDb/TMDb-style film metadata, Wikidata, and World Cities Culture Forum materials.

Values in the figures are editorial indices that structure the argument. A direct production pass should replace them with cultural trade, platform, awards, tourism, and city infrastructure aggregates.

Soft Power and Trade

Cultural exports travel alongside goods and services exports

The soft-power versus goods-power scatter places eight systems along both axes, with the strongest joint positions reaching the high 80s and 90s on the editorial scales. Culture and merchandise exports often travel together — but not always in equal proportion.

A country can export cars, chips, oil, or software. It can also export characters, songs, films, cuisine, style, and celebrity. Attention changes the world's willingness to care about the rest of the invoice.

Medium Exportability

Screen and song scale faster than place-bound culture

Film/TV scores highest on exportability at **91**, with Music at **88** and Games/anime at **84**. Sports (**80**) and Food (**77**) follow; Fashion (**71**), Design (**68**), and Literature (**63**) trail among the eight mediums.

Screen and song scale through platforms with less need for physical presence. Place-bound culture travels too, but usually needs more translation, institutions, or logistics.

Cultural Waves

South Korea accelerated, Japan compounded, and India scaled through different cultural clocks

South Korea's editorial wave index rises from **22** in 1990 to **92** by 2025 — steep acceleration. Japan compounds from an already high **71** in 1990 to **86** in 2025. India scales from **58** to **79** over the same markers.

The comparison is mechanism, not trophy: strategic acceleration, long compounding through anime/games/design, and diaspora-plus-film scale are different clocks.

Cultural Pipeline Cities

Cultural export cities combine production base with distribution reach

City pipeline scores pair production base with distribution reach. The strongest joint positions sit near the top-right of the scatter (high 90s), with a descending ladder of other export cities behind them.

Los Angeles-style production-distribution machines, Seoul-style co-ordinated export platforms, and Tokyo-style design/games/anime engines are urban launchpads for national soft power.

Cultural Trade Questions

Cultural exports ask what travels, who translates it, and what becomes identity

The diagnostic ranking elevates what becomes identity (**96**), what travels (**94**), what platform scales it (**91**), who monetizes it (**89**), what city produces it (**86**), and who translates it (**82**).

Those questions turn culture into a measurable export system without reducing it to ticket revenue alone. Attention is the cargo; identity is the durable inventory.

What this file cannot tell you

Editorial indices are scaffolding for later ingestion of UNESCO/WIPO/platform/city data — not official trade rankings. Cultural “exports” blur royalties, tourism spillovers, and platform geography.

Language markets, censorship, and diaspora size can dominate outcomes in ways a single exportability score cannot capture.

What to take away

Cultural exports belong in geoeconomics because attention moves markets: tourism, language learning, fashion, food demand, and diplomatic perception.

Screen and music travel fastest on these indices; Korea/Japan/India show different wave clocks; cities are the launchpads. Replace the indices with direct data before treating the scores as league tables.

Sources

UNESCO Institute for Statistics and UNESCO cultural trade references.

WIPO creative economy and intellectual property indicators.

World Cities Culture Forum. CREATIVE Data Framework.

MusicBrainz / MetaBrainz dumps.

IMDb/TMDb/Wikidata public metadata references.

World Bank and OECD services/export datasets.

Editor’s note

Values are editorial indices. They define the cultural export framework before direct UNESCO/WIPO/platform/city data ingestion.

Why the U.S., China, and Germany Dominate Export Debate

The United States, China, and Germany sit at the center of almost every serious conversation about trade. They are often treated as interchangeable "export superpowers" — three large economies that ship goods and services abroad at industrial scale. That shorthand collapses three different machines into one headline.

The useful comparison is not who exports more in a given year. It is what kind of global function each country performs: platform and standards power in the United States, manufacturing scale and supply-chain centrality in China, and industrial precision and machinery trust in Germany. Those roles create different kinds of leverage — and different kinds of dependence.

Start with the scale: 3 export systems compared across 7 identity axes, with 6 dependency risks scored and 8 countries placed on a substitutability map.

Research question

When policy writers call the United States, China, and Germany export superpowers, are they describing the same kind of power or three different roles inside the world economy? This report asks how platform control, factory scale, and machine-tool trust produce distinct forms of indispensability.

The observational question is comparative rather than rank-driven: which parts of global trade would be hardest to replace if one of these systems faltered, and which dependencies make each system vulnerable despite its strength? The charts use official trade agencies and editorial indices to keep those two questions separate.

Fast facts

The numbers that set the scale for this report:

3 Export superpower systems compared

7Export identity axes
6Dependency risks scored
8Countries placed on substitutability map
1Core question: who is replaceable?

Data and method

The production stack for this comparison draws on OEC and UN Comtrade product exports, partner concentration, World Bank and OECD services and GDP series, IMF macro context, and national industry statistics from agencies such as the BEA and Eurostat.

The charts here use editorial indices to define the analytical frame before a full product-level ingestion pass. They are designed to make system differences legible — not to substitute for official rankings. Observed trade totals, derived concentration metrics, and interpretive scores are kept distinct on purpose.

Three models of export power

The United States, China, and Germany represent platform, factory, and machine export systems

These countries are compared because they sit near the center of the global production system, but they do not do the same job. On the seven identity axes — scale, complexity, services and IP, manufacturing, vehicles and machinery, consumer platform power, and supply-chain centrality — each economy peaks in a different place.

The United States scores highest on services, intellectual property, and consumer platform power. China leads on manufacturing scale and supply-chain centrality. Germany concentrates strength in complex machinery, vehicles, and industrial trust. Treating them as rivals in a single market misses the architecture.

The institutional anchors are different. U.S. export power runs through firms and standards ecosystems such as Apple, Microsoft, NVIDIA, Boeing, Hollywood studios, universities, and dollar-denominated finance. China's power runs through dense manufacturing provinces, port infrastructure, electronics assembly, solar panels, batteries, and the supplier networks documented in UN Comtrade and OEC product data.

Germany's power runs through Mittelstand machinery firms, Volkswagen, Siemens, BASF, and the engineering reputation embedded in EU industrial supply chains.

Those differences explain why export volume alone is a blunt instrument. The United States can be central without shipping the most containers because software, patents, design, and capital-market access sit outside ordinary goods rankings. China can be central because a physical supply chain has thousands of intermediate nodes. Germany can be central because specialized machinery and vehicles are hard to qualify, certify, and replace quickly.

Product fingerprints

The export fingerprints show three different machines under the same global headline

A GDP total or aggregate export figure can make the three countries look like comparable blocks. The product mix — software and IP, electronics, vehicles, machinery, pharmaceuticals, energy, consumer goods — reveals three different machines under the same global headline.

Fingerprints matter because they change what a shock means. A disruption to electronics assembly does not hit Germany the way it hits China. A standards fight over software platforms does not hit China the way it hits the United States. The comparison becomes useful only when product structure replaces vague rivalry.

The United Nations Comtrade database and the Observatory of Economic Complexity both treat product categories as the basic grammar of trade. Under that grammar, a smartphone, a turbine, a car, a drug compound, a cloud subscription, and a design license are not interchangeable exports. They move through different customs codes, margins, bottlenecks, and regulatory regimes.

The product fingerprint also explains why trade disputes escalate unevenly. A tariff on finished vehicles pressures German industrial clusters; semiconductor export controls pressure China's technology upgrading path; digital-services taxes and platform regulation pressure U.S. firms whose exports are partly legal, financial, and infrastructural. The same trade headline can therefore strike three different organs of the global economy.

Dependency risk

Export strength creates different bottlenecks for each superpower

Export strength creates bottlenecks as surely as it creates leverage. The United States depends heavily on semiconductor chains, high-skill labor, and consumer and platform demand. China depends on energy inputs, shipping lanes, chips, and global buyers. Germany depends on allied markets, energy continuity, and industrial supplier networks.

The strongest systems are not invulnerable. They are specialized. Dependency scores across semiconductors, energy, consumer demand, high-skill labor, allied markets, and shipping lanes show that each superpower's risk profile tracks its product identity — not a generic trade-exposure number.

The post-2020 trade record made those vulnerabilities visible. Semiconductor shortages disrupted automobile production in Germany and the United States. Russia's 2022 invasion of Ukraine turned natural-gas exposure into a central German industrial risk. Red Sea and Suez disruptions showed how China-centered goods flows still depend on maritime chokepoints and insurance markets outside China's direct control.

For the United States, dependence often hides behind apparent strength. The country dominates platforms and advanced chips through firms such as Microsoft, Alphabet, Amazon, Apple, NVIDIA, and AMD, but those systems still lean on Taiwan Semiconductor Manufacturing Company, ASML lithography equipment from the Netherlands, high-skill immigration, and Asian electronics assembly. Platform power is not supply-chain autarky.

Who can be replaced

The strongest exporters are hard to replace for different reasons

China is hard to replace because of manufacturing scale and supply-chain integration. Germany is hard to replace because of industrial trust and precision machinery. The United States is hard to replace because of standards, platforms, capital markets, and demand pull.

The surprise is that replaceability is product-specific, not country-wide. Eight economies on the substitutability map show that the same country can look indispensable in one category and optional in another. Policy that treats China or Germany as a single switch misunderstands the map.

Vietnam, Mexico, India, South Korea, Japan, France, the Netherlands, and Taiwan can substitute for parts of the three systems, but not for all of them at once. Mexico can absorb some North American manufacturing; Vietnam can absorb some electronics and apparel assembly; India can expand services and pharmaceuticals; Japan and South Korea are critical in machinery, autos, batteries, and advanced components. None is a one-for-one replacement for China's full supplier density or Germany's high-trust industrial base.

Substitutability is also time-dependent. A buyer can reroute low-complexity consumer goods faster than it can qualify a medical-device supplier, certify a machine-tool component, or rebuild a semiconductor ecosystem. That lag is why trade resilience strategies often look expensive before a crisis and insufficient during one.

Strategic questions

The useful comparison asks who owns standards, factories, margins, shocks, allies, and substitutes

The useful comparison asks who owns standards, who controls factories, who captures margins, who absorbs shocks, who has allies, and who can be replaced. Those questions move the discussion from raw export volume into power.

Margins without factories are one kind of strength. Factories without standards are another. Allies without substitutes are a third. The chart frames the debate so that each claim about dominance has to name which of those six dimensions it means.

This framework also clarifies why "decoupling" is usually too simple a word. The United States can restrict advanced chips while remaining dependent on imported electronics. China can localize more production while remaining exposed to energy, shipping, and high-end equipment. Germany can diversify gas and export markets while remaining tied to the capital-goods cycle of Europe, China, and North America.

The better policy vocabulary is portfolio risk. Export power is a bundle of standards, factories, finance, logistics, allies, and substitution costs. A country can lead one dimension and be fragile on another, which is why the report treats superpower status as architecture rather than scoreboard position.

What to take away

Export superpowers are not rivals in one single market. They are rival system architectures — platform, factory, and machine — that create different forms of indispensability and different forms of exposure.

The best comparison asks what each country makes hard for the world to replace, and what each country secretly depends on to keep that power. Size alone does not answer either question.

Sources

Observatory of Economic Complexity. *Country and product trade profiles*. <https://oec.world/en>

United Nations Statistics Division. *UN Comtrade Database*. <https://comtradeplus.un.org/>

World Bank. *World Development Indicators: Exports of goods and services*. <https://data.worldbank.org/indicator/NE.EXP.GNFS.CD>

OECD. *Trade in Value Added (TiVA) database*. <https://www.oecd.org/sti/ind/measuring-trade-in-value-added.htm>

International Monetary Fund. *Direction of Trade Statistics*. <https://data.imf.org/?sk=9D6028D4-F14A-464C-A2F2-59B2CD424B85>

What Nations Sell—and What That Says About Identity

What a nation sells abroad is one of the few identity claims that must clear a market test. Tourism slogans can invent a self-image; export ledgers record what other countries will actually pay for.

Trade systems in the UN Comtrade and OEC style cover **200+** countries and zones and **5,000+** products under Harmonized System codes. The atlas below uses **6** diagnostic country questions and highlights **8** export families or countries to make identity readable without pretending the indices are final official rankings.

Scale and complexity are different virtues. A country can be large because it sells one thing in volume, or complex because it sells many hard-to-make things. Those identities behave differently under shock.

Fast facts

The numbers that set the scale for this report:

200+Countries/zones covered in UN Comtrade-style trade systems

5,000+Products in OEC-style trade classifications

HSHarmonized System product code architecture

6Diagnostic country questions

8Export families or countries highlighted

Data and method

The source stack includes the OEC API, UN Comtrade, World Bank WDI, OECD, IMF, and national statistics offices. OEC is particularly useful for complexity-minded views of what countries can make and sell.

These charts use editorial indices to define the comparative structure before a full product-level ingestion pass. Observed trade flows, derived complexity measures, and interpretive scores are kept conceptually separate even when they appear on the same axes.

Scale and complexity

Countries differ by export scale and product complexity

The OEC lens is powerful because exports reveal what a country can reliably make and sell into the world. Scale and complexity are different axes: volume without sophistication is one identity; sophistication without volume is another; both together is rarer.

Countries plotted high on both dimensions occupy a different strategic neighborhood from resource giants that score high on scale alone. The chart's job is to stop those neighborhoods from being collapsed into a single big-exporter label.

Product identity

Some export categories become national symbols

Cars, electronics, oil, pharmaceuticals, aircraft, software, agriculture, and textiles are not just commodities. They become reputational shorthand — Germany and machines, Saudi Arabia and energy, the United States and software services.

That is the cultural economics layer of trade: exports become part of national mythology when they are repeated long enough to feel like character. The data still matter because mythology that loses market share eventually becomes nostalgia.

The export frontier moved

The frontier moved from resources to factories to systems and IP

The last generation of globalization did not erase goods. It added services, software, intellectual property, finance, and standards to the frontier. Manufacturing indices that peak and then soften while services and IP climb are the signature of that shift.

A modern country atlas must therefore track both containers and code. An economy that only reports tonnage will misread its own power — and its own vulnerability.

Export fingerprints

The export fingerprint is the mix, not just the headline product

Germany and Japan can look similar through machines and vehicles, while Brazil and Saudi Arabia show very different resource and food signatures. The fingerprint is the mix across machines, vehicles, chemicals, resources, food, and services — not a single headline product.

The question is not only what a place sells. It is what would break if the world stopped buying it, and what prestige categories would remain even if volume fell.

Dependency questions

A country identity atlas asks what the world buys, substitutes, and depends on

The atlas ends with vulnerability questions: what does the country sell, who buys it, what inputs does it need, what could replace it, what carries prestige, and what is vulnerable? Those six prompts turn trade tables into identity analysis.

Countries that cannot answer them with data are flying blind about their own external face. Countries that answer only with slogans are doing public relations, not strategy.

What to take away

Exports are identity under constraint. They show what a country can do at scale, what the world believes it is good for, and where substitution would hurt.

Later comparison work can rank places by complexity, concentration, cultural export power, and substitutability — but only after the fingerprint and the six diagnostic questions are in view.

Sources

New York and San Francisco: Command City vs. Invention City

New York and San Francisco are not peers by population or geography, but they are peers in the market for invisible power. They both export coordination, talent, symbols, and future-facing industries.

The question is why they belong in the same conversation and what people miss when they compare them only by rent, taxes, or office vacancy.

Start with the scale: **2** — Global intangible-output cities compared; and **8** — Identity axes scored.

Fast facts

The numbers that set the scale for this report:

2Global intangible-output cities compared

8Identity axes scored

6Shared pressure narratives tested

5Charts in this comparison

0Single-variable answers

1Core contrast: command versus invention

Data and method

The source stack includes BEA metro GDP, Census ACS, BLS employment, DataSF, NYC Open Data, World Cities Culture Forum, public venture-capital references, transit agencies, and office-market summaries.

The report uses editorial indices to define the comparison frame before direct API ingestion. It also references the San Francisco Data Microscope as the local profile layer.

Command versus Lab

New York commands capital and culture while San Francisco prototypes new systems

New York and San Francisco are compared because both export invisible value: finance, media, software, talent, coordination, status, and ideas.

The difference is posture. New York is the command center; San Francisco is the laboratory.

History Timing

New York compounds older command layers while San Francisco spikes around paradigm shifts

New York's power is layered and old: port, finance, media, corporate headquarters, immigration, culture. San Francisco's modern power is sharper and more episodic: software, venture capital, platforms, AI.

That is why both cities feel globally central while operating on different clocks.

Output Fingerprint

The cities both export intangible systems but not the same systems

New York exports capital allocation, status, media, fashion, finance, and market legitimacy. San Francisco exports software platforms, venture risk, AI, and a cultural permission structure for invention.

The pairing is valid because both cities sell coordination more than physical goods.

Pressure Pairing

Post-2020 pressures rhyme across the two cities but come from different causes

Both cities talk about office vacancy, housing, disorder narratives, transit, and talent churn. But the same words are not the same system.

New York's office reset sits inside a giant command economy. San Francisco's sits inside a smaller city whose downtown was overexposed to tech-office rhythms.

What to Ask Next

The useful comparison asks which city commands, invents, absorbs, builds, renews, and exports status

The surprise is that neither city simply beats the other. The answer changes by output: capital, platforms, culture, housing, talent, or status.

This gives future city reports a linked framework instead of isolated profiles.

What to take away

The central finding is that New York and San Francisco are comparable because both sell systems more than goods. The difference is that New York legitimizes and commands while San Francisco invents and prototypes.

That means the richer comparison is not which city is better, but which system each city is best at producing.

Sources

BEA. Metropolitan GDP and regional industry data.

U.S. Census ACS.

BLS metro employment data.

DataSF and NYC Open Data Socrata portals.

World Cities Culture Forum CREATIVE Data Framework.

Transit agency and office-market public summaries.

Editor's note

Values are editorial indices designed to structure a later direct-data comparison. They should be replaced with source aggregates for formal ranking.

California and Texas: Where Rival State Systems Diverge

California and Texas are the most common American state comparison because each is large enough to stand in for a theory of the country. One is cast as the expensive knowledge-and-culture machine; the other as the growth-and-capacity machine. The framing here scores 2 state systems across 7 rivalry dimensions and 7 export/industry families.

The charts use editorial indices to structure the argument before a full direct-data production pass. They are designed to stop the rivalry from collapsing into a single tax talking point — and to surface the six comparison questions worth asking next.

Fast facts

The numbers that set the scale for this report:

2State systems compared
7Rivalry dimensions scored
7Export/industry families compared
6Next comparison questions
0Simple answers accepted

Data and method

A full production stack for this comparison includes BEA state GDP by industry, Census ACS population and migration, IRS migration summaries, BLS employment, state tax and budget documents, OEC/ITA export tables, and climate/energy series.

Values in the figures are editorial indices that organize the rivalry into comparable dimensions. They should be read as a structured hypothesis map, then replaced with source aggregates before formal rankings.

Why They Are Compared

California and Texas are rival systems because both are big enough to feel like countries

On the rivalry radar, California scores at the top on GDP scale (**96**), population (**94**), tech role (**94**), and cultural output (**91**), with a high tax-burden debate score (**88**) and a low energy-role score (**46**). Texas mirrors with energy at **96**, migration attention at **91**, population at **88**, and GDP scale at **86**, while cultural output sits lower at **68**.

People compare them because they are competing governance models, labor markets, export systems, and migration symbols — not because they are interchangeable states with different logos.

Growth Model

California compounds knowledge while Texas compounds growth capacity

California's knowledge/IP model index stays high across the timeline — from the low **90s** in 2000–2010 to a peak near **98** in 2020. Texas's growth/energy model climbs from about **61** in 2000 to about **90** by 2025 in this editorial series.

One compounds intellectual property, entertainment, venture networks, and universities. The other compounds energy, land, logistics, and relocation capacity. The rivalry persists because the models are not substitutes.

Export Fingerprint

The states overlap in scale but diverge in what they export to the world

The export fingerprint contrasts software/IP, entertainment, and related strengths on the California side with energy and logistics weight on the Texas side, while agriculture, aerospace, and other families overlap in more complicated ways than slogan versions admit.

Same-scale GDP conversations hide different machinery. What each state sells to the world is the cleanest way to keep the comparison honest.

Hidden Tradeoffs

The rivalry is about cost, climate, wages, culture, and risk, not taxes alone

Cost, climate risk, wages, culture, and business formation do not move as a single tax variable. The tradeoff scatter keeps multiple dimensions in view at once — high-cost magnetism on one side, capacity and exposure on the other.

California's wage ceiling and cultural pull are real; Texas's formation and logistics advantages are real; energy and climate risk are real. A serious comparison refuses to delete any of those axes.

What to Ask Next

The useful California-Texas questions are about irreplaceable exports, talent, climate, and who benefits

The ranked questions elevate talent compounding (95), irreplaceable exports (93), climate handling (91), who benefits from growth (88), whether migration is tax-driven (86), and what each state must import (79).

The useful prompt is not “which state is better?” It is which capabilities each system cannot copy from the other — the starting point for migration, housing, export, and culture follow-ups.

What this file cannot tell you

Editorial indices are scaffolding, not BEA or Census tabulations. They must be replaced with source aggregates before anyone treats the scores as official rankings.

State borders hide metro heterogeneity: coastal California is not inland California; Houston is not the Rio Grande Valley. Intra-state variance can exceed the interstate gap on many metrics.

What to take away

California and Texas remain comparable because they are two large answers to how America organizes scale, talent, land, energy, and culture.

Taxes are the entry point; export identity, climate risk, talent compounding, and cultural magnetism are where the comparison becomes specific enough to measure.

Sources

BEA. State GDP by industry.

U.S. Census ACS and population estimates.

IRS migration data summaries.

BLS state employment statistics.

International Trade Administration and OEC export references.

State budget and tax agency documents.

Editor's note

Values are editorial indices intended to structure a later direct-data production pass. They should be replaced with source aggregates before making formal rankings.

Cities as Operating Systems of Exports, Culture, and Scarcity

A city report should not start with a postcard. It should start with a system map: what the city makes, what it imports, who it serves, what it prices out, what it remembers, and what it cannot replace.

this piece creates the frame for the next geo-economics layer of Arto-metrics. Comparisons can come later; first the site needs a way to ask the right questions of a place.

Start with the scale: **1,667** — Subnational regions in DOSE v2.14 public documentation; and **45** — Cities in World Cities Culture Forum 5th edition summaries.

Fast facts

The numbers that set the scale for this report:

1,667 Subnational regions in DOSE v2.14 public documentation
45 Cities in World Cities Culture Forum 5th edition summaries
1,100+ Datasets in DataSF public portal summaries
6 Diagnostic questions in the city microscope
8 System layers scored

Data and method

The source stack includes BEA regional GDP, Commerce metro exports, DOSE global subnational output, Census ACS, World Cities Culture Forum, and local Socrata portals such as DataSF or NYC Open Data.

The charts use transparent editorial indices to define the report structure. A production pass should replace each index with a direct API or CSV aggregate.

System Layers

A city identity is a stack of economic and cultural systems

A city is not one industry. It is a layered machine: firms, ports, universities, housing, cultural venues, finance, infrastructure, rules, and stories.

The first Artometrics question is not whether a city is good. It is what kind of system the city is trying to be.

Goods and Services

Some cities export goods while others export coordination, code, finance, or attention

Houston and Los Angeles still have obvious goods/logistics signatures. New York and San Francisco sell more invisible products: finance, software, expertise, media, and market access.

That matters because invisible exports are harder for casual viewers to see, but they often define the city's price structure.

History Layers

Ports, factories, corporate headquarters, and software leave measurable shadows

History is not background flavor. It is infrastructure that keeps generating data: street grids, warehouses, universities, firm clusters, zoning, and habits of migration.

A city report should explain which older layer is still driving the present numbers.

City Fingerprint

The city fingerprint appears across GDP, housing, talent, culture, and export complexity

One variable lies. A fingerprint requires a pattern. The same GDP level can mean very different things under different housing, transit, culture, and export structures.

This is why a city profile should read like a data portrait, not a fact sheet.

Questions

The city microscope starts with what the place makes, imports, lacks, and competes against

Before building comparisons, the local report needs its diagnostic questions. What does the place make? What does it import? What is scarce? Which cities are its real competitors?

Those questions turn open data into an identity argument.

What to take away

The central assumption is that cities have data signatures. The challenge is to separate slogan from system: what is produced, what is scarce, who competes, and what historical layer still controls the present.

This gives future city reports a repeatable spine before deeper city-to-city comparisons begin.

Sources

BEA. Regional GDP by county and metropolitan area documentation/API.

DOSE / PIK. Global subnational economic output dataset.

World Cities Culture Forum. CREATIVE Data Framework and 5th Edition Data Explorer.

DataSF and NYC Open Data. Socrata/SODA portal documentation.

U.S. Census ACS and International Trade Administration Metropolitan Export Series.

Editor's note

Values are editorial indices designed to define the analysis contract. They should be replaced with direct source aggregates during city-specific production passes.

Which National Parks Draw the Crowds?

National parks are sold as wilderness and remembered as crowds. Visit counts make that tension measurable: which named units absorb America's outdoor attention, and how that attention has compounded since the early twentieth century.

A TidyTuesday working file of **21,560** records spans **1904–2016**. Median visitors sit near **155,219**; the highest observed visitors figure reaches **871,922,828** in the extreme cell of the extract. **Golden Gate** leads the park-name ranking used in the fact box, and region code **IM** is the most common regional label.

Urban gateways and memorial parkways often outdraw remote iconic wilderness. The leaderboard is as much about access and metropolitan adjacency as about scenic purity.

Fast facts

The numbers that set the scale for this report:

21,560Records in the working dataset
155,219Median Visitors
871,922,828Highest observed Visitors
Golden GateTop Parkname by Visitors
1904–2016Year span covered in the file
IMMost common Region

Data and method

The source is the TidyTuesday release from 2019-09-17 (R for Data Science community). The working file contains 21,560 rows and 13 columns after assembly — park names, regions, years, and visitor totals among them.

Medians handle skew from a few hyper-visited units. Charts export as Plotly JSON with PNG fallbacks. Visitor totals are administrative counts with known definition changes over a century; treat long-run levels as directional, not penny-precise.

How visits grew over time

Median Visitors Over Time

Median visitors rise from about **2,200** in the opening period to about **198,478** at the close — nearly two orders of magnitude of growth in the typical unit's recorded traffic.

That climb tracks automobile access, population growth, the expansion of the park system itself, and the conversion of outdoor recreation into a mass middle-class habit. The median's rise is the system story; the leaders are the celebrity story.

Who sits at the top

Golden Gate leads at 14,554,750 — 5,151,270 marks the median among the top dozen

Golden Gate leads at about **14,554,750** on the highlighted ranking, with **5,151,270** as the median among the top dozen. Related high-traffic units in the broader leader cuts include urban memorials and parkways — Vietnam Veterans Memorial, World War II Memorial, Lake Mead, Natchez Trace, Cape Cod — places where recreation, commuting adjacency, and civic pilgrimage overlap.

Wilderness icons still matter culturally. They do not always win the raw visit count contest against metropolitan National Park Service units that function as public squares with better transit.

How regions spread visits

Visitors by Region

Regional box plots show whether visitor consensus is shared or contested across NPS regions. Some regions host many modest units; others host a few magnets that dominate the regional total.

Spread inside a region can exceed differences between region medians. A single gateway park can outdraw an entire cluster of remote units.

Who beats the median — and who trails

Visitors vs median by Region

On the gap chart, region NT sits about **12,634,481** above the median; **AK** trails by about **142,104**. Those gaps encode access and population gravity as much as scenic ranking — Alaska's vast parklands are not built for casual weekend volume.

Reading the gap as a quality score would insult both the crowded memorials and the remote wilderness. It is a map of visitation pressure, not of beauty.

Concentration of visits

The top 5 parkname entries account for 52% of the aggregate visitors

The top five park-name entries account for **52%** of aggregate visitors in the working cut. A slight majority of the system's recorded traffic concentrates in a tiny head of names.

Steep concentration means infrastructure stress, concession economics, and reservation politics will always be about a short list of places — even while the agency's mission statement names a much longer list.

What this file cannot tell you

Community-cleaned TidyTuesday snapshots are not live APIs. Missing values, spelling variants, and century-scale definition changes apply. Visitor counts are not unique individuals, and they are not measures of ecological impact by themselves.

Findings describe structural signals about National Park Service visitation patterns in the file — not a complete conservation audit or a ranking of which landscapes matter most.

What to take away

Park visitation is a growth story with a concentrated head: median traffic rose from thousands to near 200,000, while a handful of units still absorb roughly half of aggregate visits.

The citable tension is access versus myth. America's most visited park units are often the ones closest to cities and memorials, not only the ones on the postcard wilderness calendar.

Sources

San Francisco as an Invention Machine Under Housing Pressure

San Francisco is a perfect city microscope because its contradictions are measurable: a small city with enormous software output, extreme housing pressure, intense cultural mythology, and a regional economy larger than its municipal boundary.

this piece does not try to finish the comparison yet. It defines the questions a serious SF data portrait has to ask.

Start with the scale: **1,100+** — DataSF public portal dataset scale in public summaries; and **7** — Identity domains scored.

Fast facts

The numbers that set the scale for this report:

1,100+DataSF public portal dataset scale in public summaries

7Identity domains scored

6Open questions for the next SF production pass

3Core tradeoff signals: tech, housing, tolerance

6Competitor cities framed

Data and method

DataSF provides Socrata/SODA API access to city datasets, including registered businesses, building permits, 311 cases, public safety, transportation, housing, and civic operations. Regional identity requires adding BEA, Census ACS, BLS, Commerce export data, and Bay Area transportation sources.

These charts are a framework pass: they show what the SF report should test when direct APIs are pulled.

Identity Stack

San Francisco exports software, finance coordination, and cultural permission

The simplest wrong answer is that San Francisco exports apps. The deeper answer is that it exports systems: venture risk, engineering labor, platform logic, taste, and institutional permission for strange ideas.

That is why the city can be small in population but enormous in economic imagination.

Tradeoff History

Technology concentration rose faster than the city could absorb housing pressure

The SF story is not only success. It is absorption capacity. A small peninsula city generated global software wealth while underbuilding the physical city needed to metabolize that wealth.

The assumption to test later is whether housing, downtown vacancy, transit, and cultural retention are symptoms of the same bottleneck.

Neighborhood Pressure

Culture and cost rise unevenly across the neighborhood map

A citywide average misses the point. SoMa, the Mission, Chinatown, the Tenderloin, and the Sunset each carry a different mixture of price, memory, commerce, and political meaning.

The DataSF layer matters because the city is small enough for block-level data to change the argument.

Competitor Set

San Francisco competes against other talent ecosystems, not generic cities

Seattle is a software competitor. New York is a capital and talent competitor. Boston is a research competitor. Austin is a cost/relocation competitor.

The right competitor depends on the variable being contested.

Open Questions

The SF data story is about whether the city can absorb its own invention machine

The next report should not merely rank San Francisco. It should ask whether the city can build housing, refill downtown, retain artists, keep AI local, and connect biotech to the same regional engine.

That is the bioeconomics question: can a place survive the metabolism of its own strongest trait?

What to take away

The main claim is that San Francisco's identity is an invention machine constrained by physical absorption. The city makes ideas, companies, and cultural permission faster than it makes space.

That claim can later be tested against building permits, business registrations, office vacancy, transit recovery, migration, venture capital, payroll, and cultural venue data.

Sources

DataSF / SF.gov. Open Data portal and developer documentation.

Socrata / SODA API documentation.

BEA. Metropolitan GDP and regional data.

U.S. Census ACS. Housing, income, commuting, and demographic tables.

International Trade Administration. Metropolitan Export Series.

World Cities Culture Forum. CREATIVE Data Framework.

Editor's note

Values are editorial indices for a source-backed framework. They should be replaced with direct DataSF, BEA, Census, BLS, and export aggregates in the full city-specific report.